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A Project Report on

"CIAO - Code In Art Out"

[Code No: COMP 206]

(For partial fulfillment of 3rd Semester in Computer Science)

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Bona fide Certificate

This project work on

“CIAO - Code In Art Out”

is the bonafide work of

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who carried out the project work under my supervision.

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Date: 23rd Nov 2022

Abstract

In recent years, computer programming has become an increasingly popular way to create art. Using computer programming, everything from mathematical patterns like fractals, to biological ones like phyllotaxis, or even something very abstract can be rendered. We have created "CIAO - Code In Art Out", a web app that enables users to produce generative art of different kinds. Generative art refers to computer generated arts which are rendered using different sets of algorithms. CIAO allows its users to create generative art in one of 2 ways: first by uploading an image to transform into generative art, or second by selecting a template from a grid of options which users can interact with to create their own version of the art. It is developed using react and p5.js which are javascript frameworks. MongoDB is used as a database to store user's favorites and their created art while Node.js and Express.js is used to handle the backend. CIAO is primarily intended to educate users on generative art algorithms. It also aims to boost creativity by allowing users to play with an infinite amount of ideas to produce everything from something very abstract and simply to extremely complex and detailed.

Keywords: Generative art, JavaScript, React, MongoDB, p5.js, Node.js, Express.js

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Acronyms/Abbreviations

CIAO:	Code In Art Out
HTML:	HyperText Markup Language
CSS:	Cascading Style Sheets
MERN:	MongoDB, Express, React and Node
CRUD:	Create, Read, Update and Delete

Chapter 1: Introduction

1.1 Background

Since the beginning of time, there have been many great artists ranging from Leonardo Da Vinci, the creator of the most iconic painting, Mona Lisa to J.K. Rowling, the writer of one of the most famous fantasy novels, Harry Potter. Several other famous pieces of art such as The Starry Night from Vincent van Gogh, David from Michelangelo, Für Elise by Beethoven, The Godfather directed by Francis Ford Copolla and many others have shaped human history. Art has evolved over time and a relatively new approach to making art is the use of Computer programming. This kind of art is called generative art.

Generative art refers to any art practice where the artist uses a system, such as a set of natural language rules, a computer program, a machine, or other procedural invention, which is set into motion with some degree of autonomy contributing to or resulting in a completed work of art. [1] Generative art can be everything ranging from paintings to poems and stories. However, this project will be specifically designed to create generative paintings. Programmers can use knowledge of math and science to generate everything from highly detailed art to something very abstract in a canvas. Inspiration can also be drawn from nature. For instance, the mandelbrot set (Set of complex numbers c for which the function $f_c = z^2 + c$ does not diverge to infinity when iterated from $z = 0$, i.e, for which the sequence $f_c(0)$, $f_c(f_c(0))$, etc remains bounded in absolute value [2]) when visualized generates a beautiful fractal while an algorithm for generating phyllotaxis, the regular arrangement of lateral organs (leaves on a stem, scales on a cone axis, florets in a composite flower head [3]) creates a mesmerizing pattern.

With our web app CIAO, Users can create generative art in one of two ways. The first is by selecting from a grid of options of generative art which can be individually altered by using color pickers and sliders to adjust to the value of variables to get different results. Furthermore, the user can also upload an image to create their own generative art from their chosen image. The app can be used with or without logging in. However, registering as a user comes with various additional features like downloading the generated art, saving art to favorites, and uploading the user's art to a database.

1.2 Objectives

- Increase creativity by experimenting with a countless number of ideas to create everything from very precise art to something very chaotic and random.
- To better grasp the subject, visualize numerous mathematical and science topics.
- Create eye-catching artwork from templates or pictures that may be used as everything from wallpaper to NFTs.
- Introduce users to generative art algorithms and patterns in nature.

1.3 Motivation and Significance

Some people might be good at art while some people might be good at programming. With CIAO, we aim to narrow the gap between art and technology. CIAO makes it possible for everyone to be an artist. Generative art can also be a great way of learning about programming concepts while also being able to create something that is visually appealing. Furthermore, Creating art can be a relaxing experience. Art and drawing are known to be great methods to help to reduce stress. In fact, there is an entire field called art therapy. [4] Not only will this be a means of recreation but it will also aid in visualization of different concepts of mathematics and science.

Chapter 2: Related Works/ Existing Works

Generative art is an emerging form of art that uses mathematics to create art from code. There aren't many digital platforms that create generative art. Some of the online web applications are Acrylicode, Ascii Art Generator, Silk, Chromata , etc.

2.1 Repper

Repper [4] is a user-friendly paid web application that creates beautiful geometric patterns seamlessly that are useful for web design and print on demand products.

Features:

- It allows us to work from a base image that can be imported from our file or unsplash to create generative art.
- We can see how the controls change or affect your art or pattern in real time.
- It allows us to add different pattern and effect like color adjustment, gradient, metamorphosis, rotation, etc

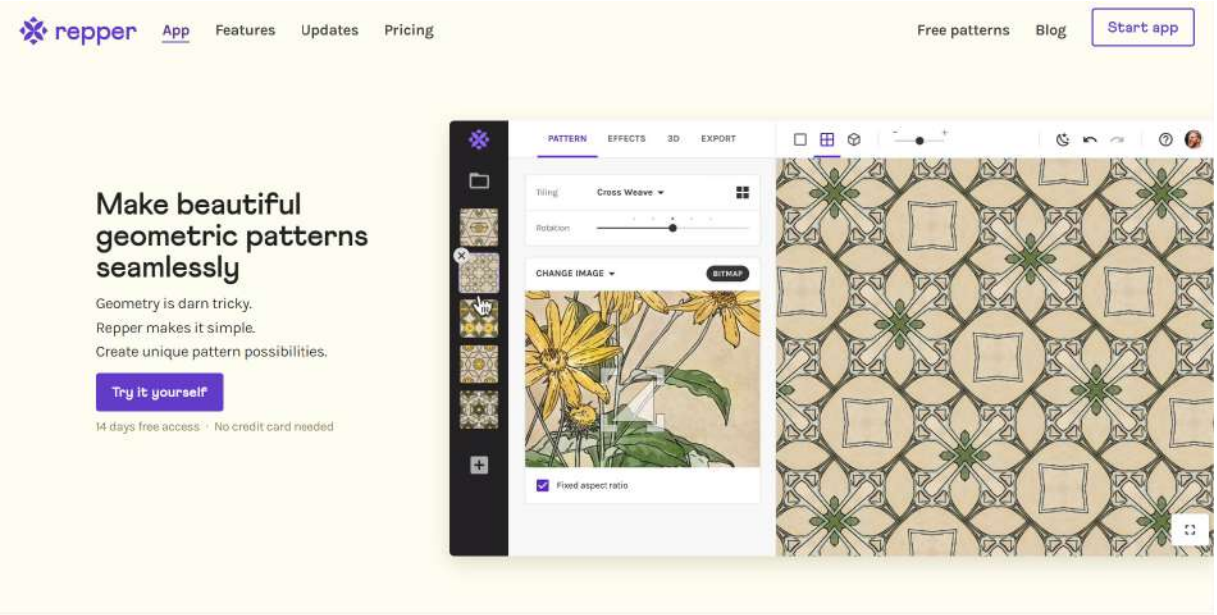

Fig 2.1.1: Repper.app

2.2 Acrylicode

Acrylicode [5] is a web application which automatically creates generative art using a drawing harmonization tool .

Features :

- It allows us to draw any line / shapes /pattern on a screen to create generative art by varying rotating angle, number of shapes, scaling factor and zoom.
- We can save our generated art in svg file type .

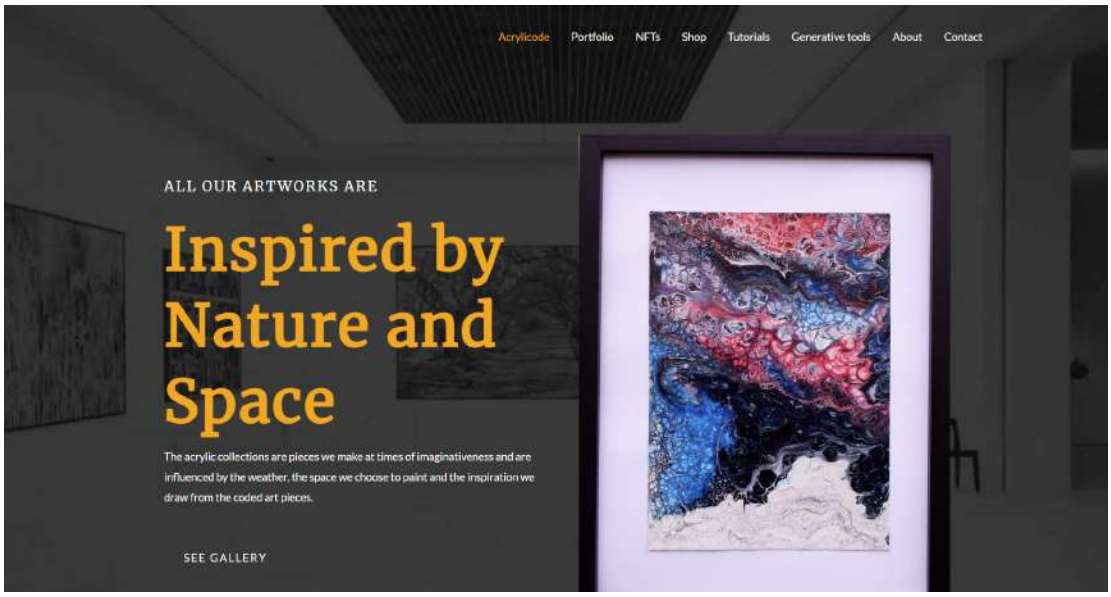

Fig 2.2.1: acrylicode.com

2.3 Ascii Art Generator

Ascii Art Generator [6] uses code to generate images created by converting an existing image to Ascii characters.

Features :

- It allows us to upload and convert:
    1. Image to Monochrome Ascii Art
    2. Image to Color Ascii Art
    3. Text to Ascii Art Banner
- It supports variety file types : jpg, png, bmp, gif, jpeg,etc
- All supported files are deleted automatically after half an hour for user privacy.

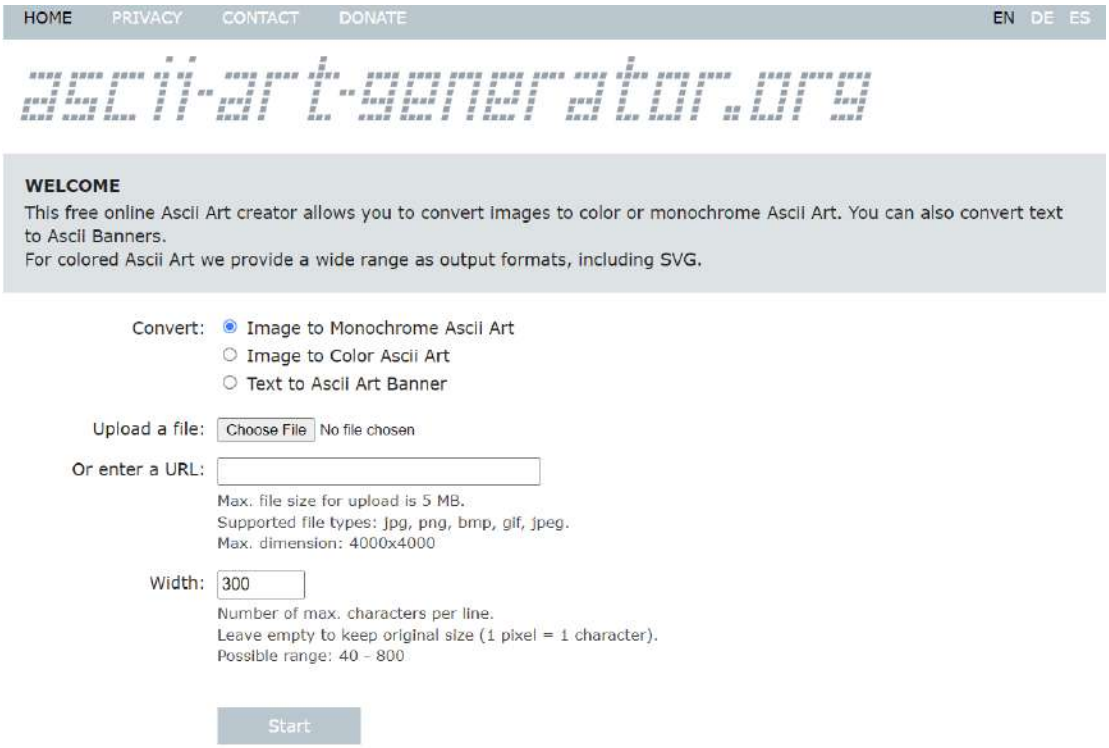

Fig 2.3.1: ascii-art-generator.org

2.4 Silk

Silk [7] is an online tool for creating symmetrical art, including rotational symmetry. It is also available as an app to create impressive designs with a swipe of a finger.

Silk's interactive nature is a significant model of how simple rules such as including symmetrical mirroring of the lines, can instantly transform basic input into much more detailed, complex and advanced forms.

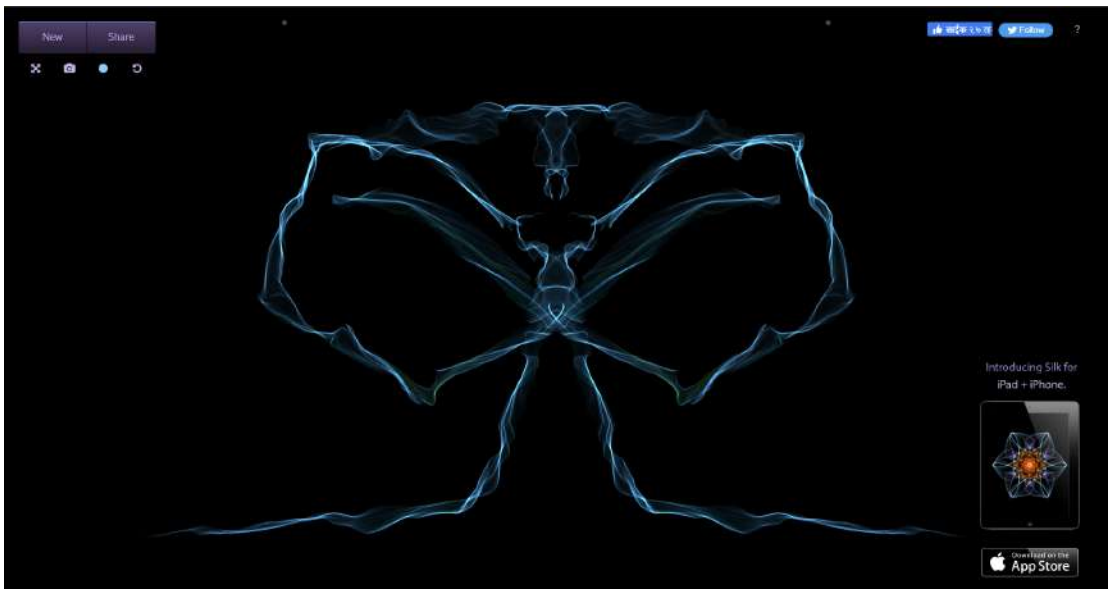

Fig 2.4.1: weavesilk.com

It also creates impressive designs with a swipe of a finger. A significant model of how simple rules such as including symmetrical mirroring of the lines, can instantly transform basic input into much more detailed, complex and advanced forms is created by snowflakes.

Chapter 3: Design and Implementation

3.1 System Requirement Specifications

3.1.1 Software Specifications

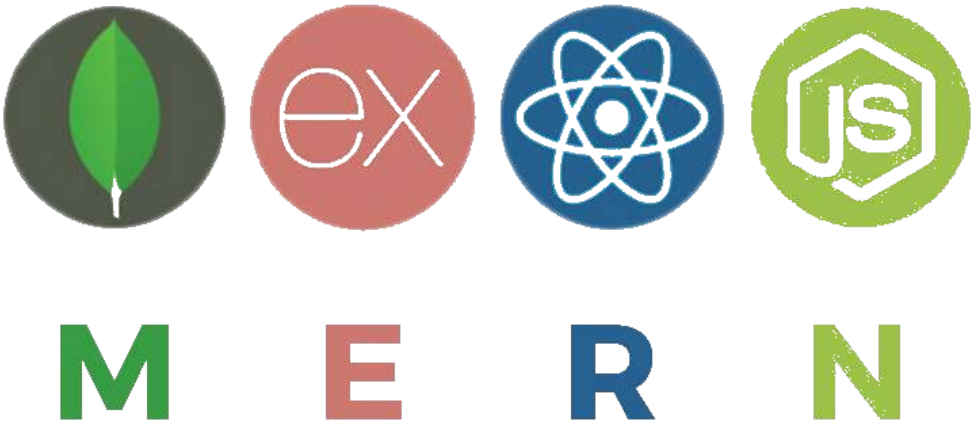

Fig 3.1.1: MERN Stack

This web app uses the MERN stack for development. It stands for MongoDB, Express.js, React.js and Node.js. Furthermore, it uses p5.js javascript library for generating the art.

3.1.1.1 Front End Tools

React.js: React is a frontend JavaScript library for building fast and interactive UI components. React can be used as a base in the development of single page applications. React is one of the most popular JavaScript libraries used for building UI.

Bootstrap: Bootstrap is a frontend CSS framework for faster and easier web development. It helps in creating fully responsive layouts by providing templates that use predefined css classes.

Material UI: Material UI or MUI is a comprehensive library that provides pre-built components implementing Google's material design system. It drastically reduces development time by providing often used components like alerts, backdrop, buttons, dialog box and accordion.

p5.js: p5.js is a free and open-source JavaScript library for creative coding. It has a full set of tools that simplify creating interactive visuals with code that works right in a browser.

3.1.1.2 Back End Tools

MongoDB: MongoDB is a leading NoSQL database and an open-source document database. A lot of big production companies work with MongoDB. MongoDB is a document database that is intended to be simple to construct and design.

Node.js and Express.js: Node.js is a javascript runtime environment that is used to build server side applications while Express.js is a server-side framework that runs inside a node.js server. This combination allows us to perform CRUD operations on the database. Other libraries like bcrypt for password hashing, validator for validation of email and password format, JWT for user authentication have also been used.

3.1.2 Hardware Specifications

This project is a web application, so any sort of sophisticated hardware is not required. It can function as long as there is a working internet connection.

3.2 Project Structure

CIAO uses a combination of react with p5.js to create appealing and dynamic generative art that can be highly customized to the user's preference. Each art in the project acts as its own dependent child component. The parent component contains sliders and color pickers for customizing the art. They have a default value which is passed as prop to the child art component to render the art. Furthermore, the art component is designed such that every time the value of the slider and color picker changes which in turn changes the props, the sketch rerenders with the new value of the provided props.

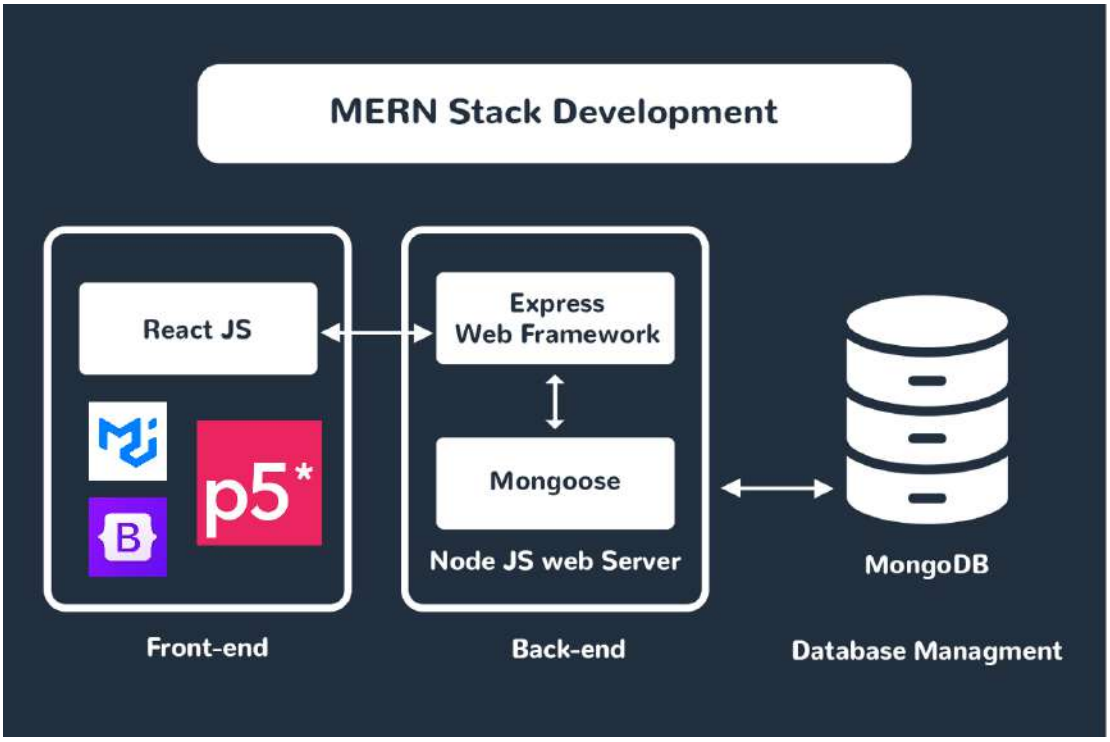

Fig 3.2.1: Project structure

As for the additional features that come with logging in, the favorites are stored as an state object with key, value pairs. The key is the unique id of the art while the value is a boolean which signifies whether the art is a favorite or not. When required, this information is queried in the database to display only the art set as favorite.

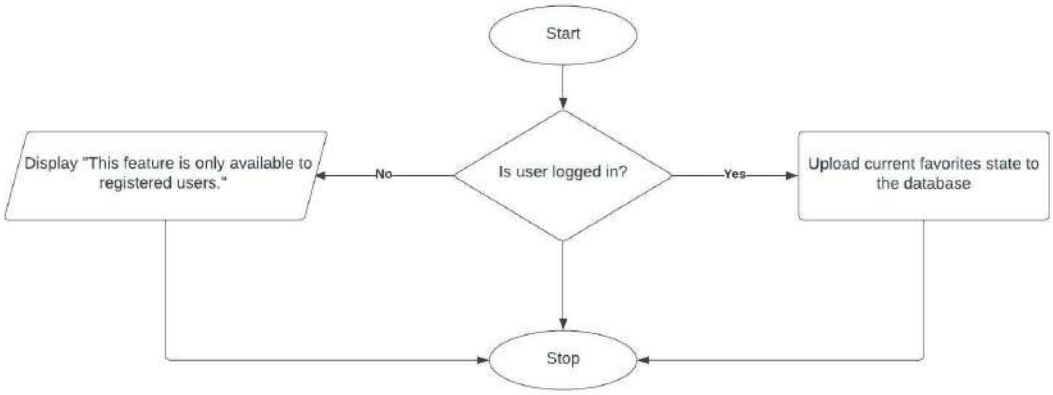

Fig 3.2.2: Flowchart for adding art to favorites

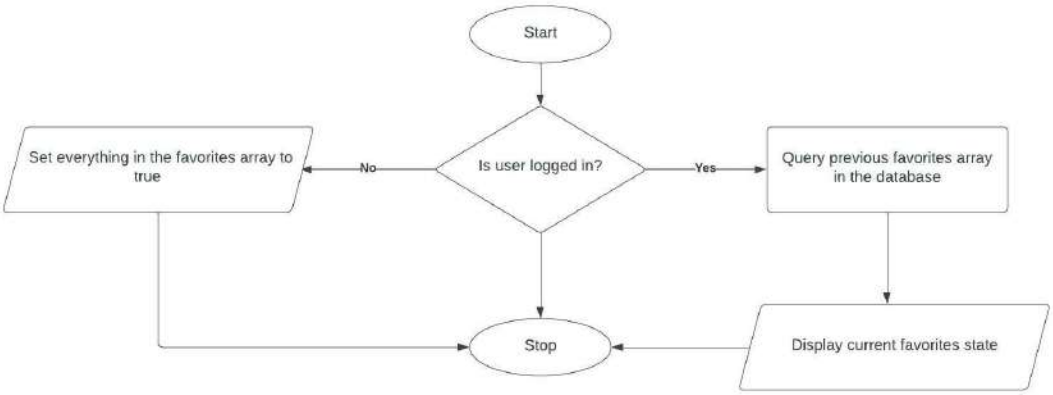

Fig 3.2.3: Flowchart for viewing favorites

```
favorites: Object
  1: false
  2: false
  3: false
  4: true
  5: true
  6: false
  7: false
  8: false
  9: false
  10: false
  11: false
  12: false
  13: false
  14: true
  15: false
  16: false
  17: false
  18: true
  19: false
```

```
arts: Array
 ⌄ 0: Object
     branchlength: 69
   › leafcolor: Object
   ⌄ backgroundcolor: Object
     › rgb: Object
   ⌄ trunkcolor: Object
     › rgb: Object
   ⌄ resolution: Object
       x: 900
       y: 650
     id: 5
 ⌄ 1: Object
     increment2d: 7
     bold2d: 1.4
     layers: 24
   ⌄ backgroundcolor: Object
     › rgb: Object
   ⌄ strokecolor: Object
     › rgb: Object
     id: 2
   ⌄ resolution: Object
       x: 300
       y: 300
```

Fig 3.2.4: Favorites Object and Arts array

Moreover, the saved art feature works not by storing the canvas as an image but as an object which consists of the selected value from sliders and color pickers. The saved arts exist as an array of these objects in the database. When the user wishes to view their saved art, the information from this array is used to render the sketch.

Upon opening the website, the user is presented to a home screen. The user can navigate around the app with the navbar. When the user registers, they are redirected to the homepage. They can also navigate around the app but have restricted features like saving and downloading the templates. But if the user is logged in, they are not restricted to these features. The user can also upload their own images and add effects to them with the FromImage page and download the generated images.

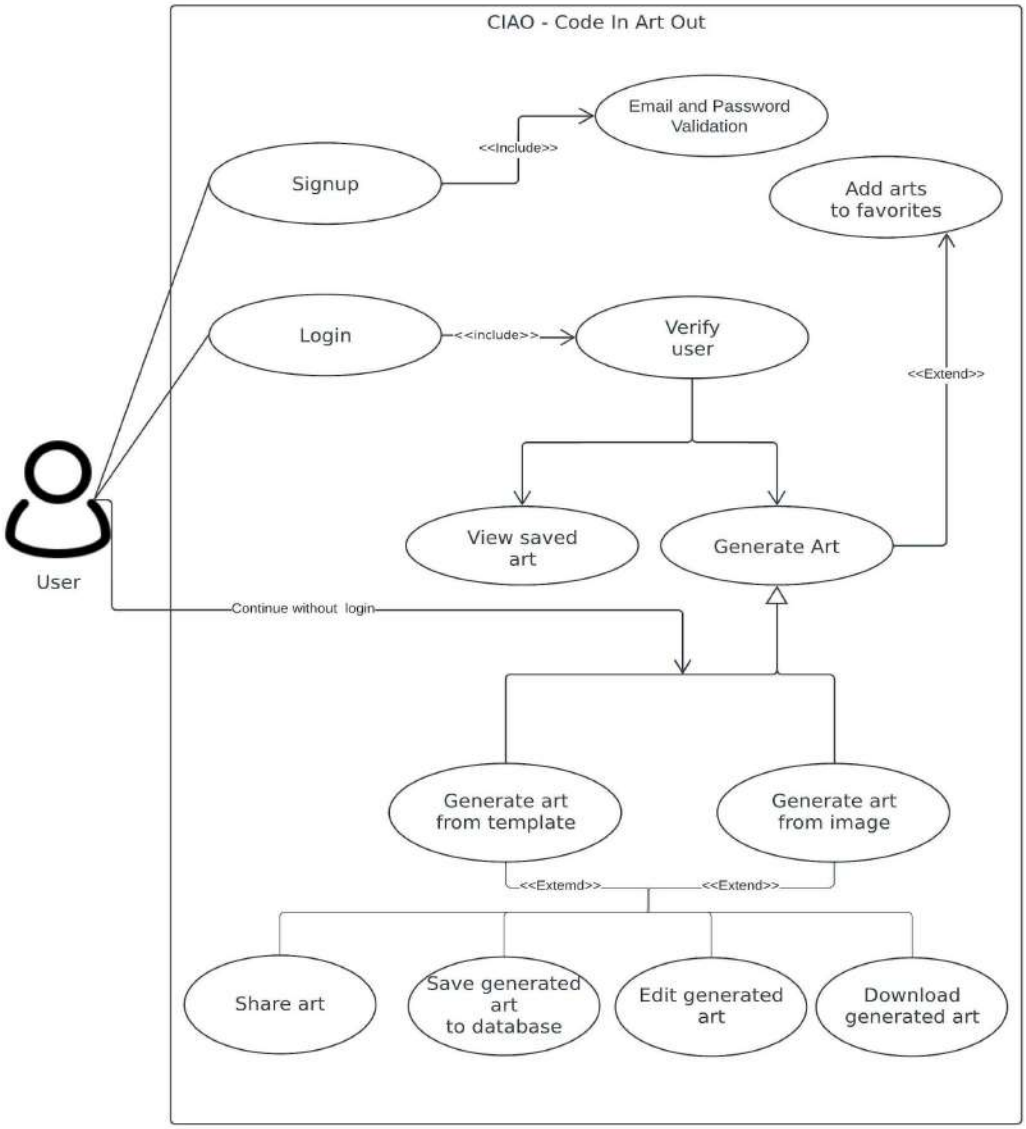

Fig 3.2.5: Use Case Diagram

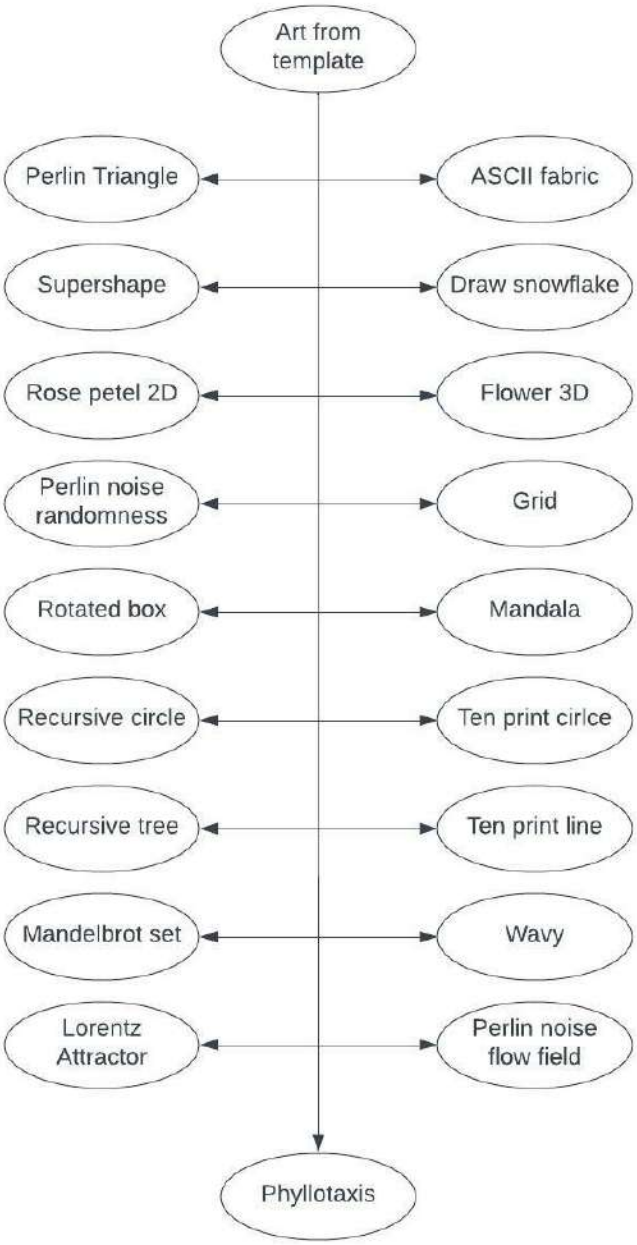

Fig 3.2.6 : Art from template

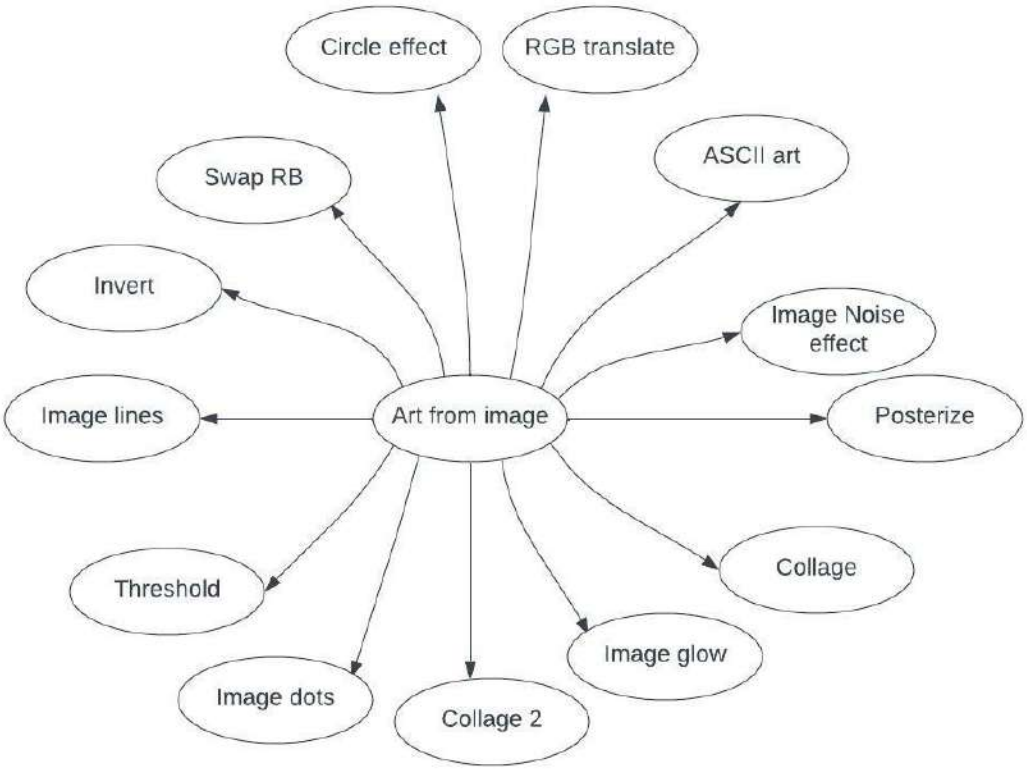

Fig 3.2.7: Art from image

Chapter 4: Discussion on the achievements

4.1 Features

4.1.1 Generate art from template

The user can generate art by choosing from 19 different templates including Phyllotaxis, Perlin Noise flow field, Lorentz Attractor, MandelBrot Set, Recursive tree, Recursive circle, Ten Print line, Ten Print Circle, Rotated box, Wavy, Grid, Mandala, Perlin Noise Randomness, ASCII Fabric, Rose Petal 2D, Supershape, Perlin Triangle, Flower 3D, Draw Snowflake.

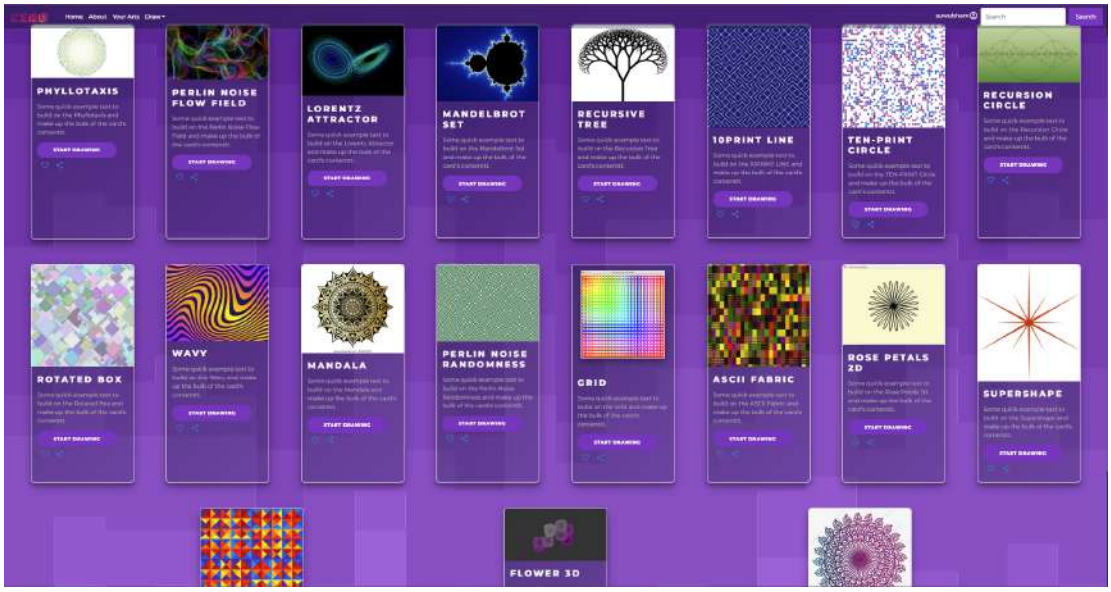

Fig 4.1.1.1: Generate art from template (zoomed out)

4.1.2 Edit customized art

The user can change the generated art to their preference by using sliders and colorpickers. Each art has been carefully designed to be able to make thousands of possible results on changing the value of the sliders and color pickers. For instance, In Recursive tree, everything from the resolution, branch length to the leaf color, background color and trunk color can be changed.

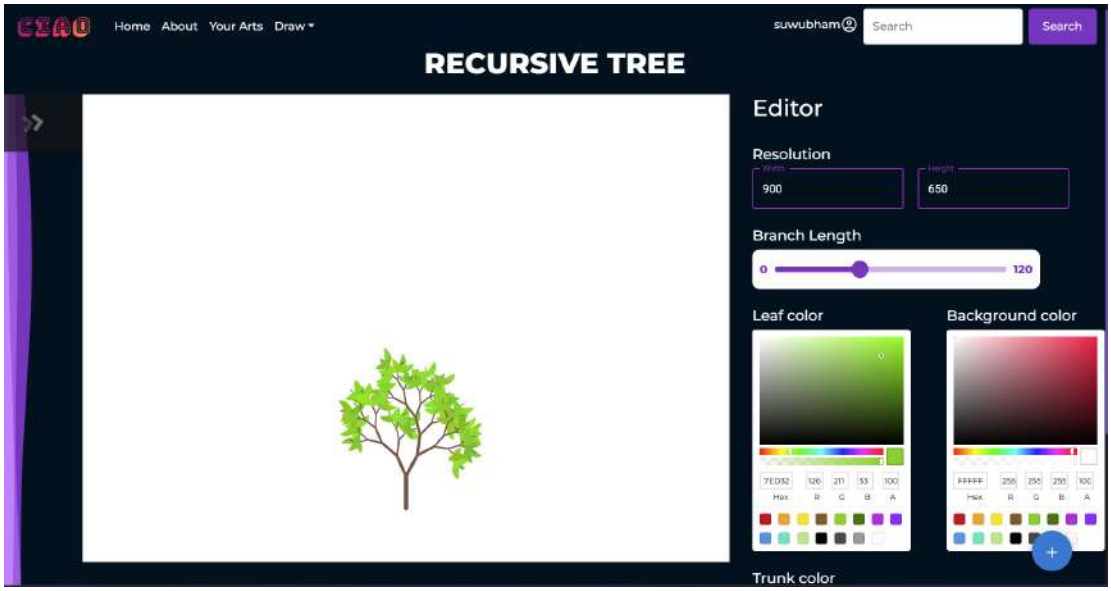

Fig 4.1.2.1 : Before customizing generated art

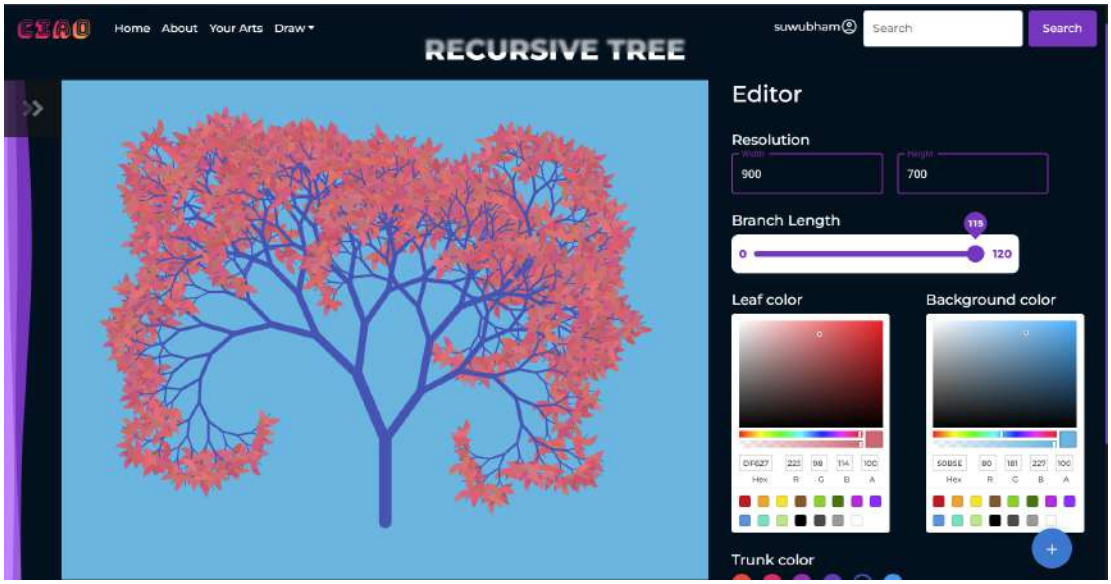

Fig 4.1.2.2: After customizing generative art

4.1.3 Adding effect on the picture

The user can alter their images using one of the 13 new additions to the art from image effect. The user can choose from 13 distinct effects to convert their art into images. We have effects such as Circle effect, RGB Translate, ASCII Art, Image noise, Posterize,Collage, Image Glow, Imagedots, Threshold, Imagelines, Invert and SwapRB.

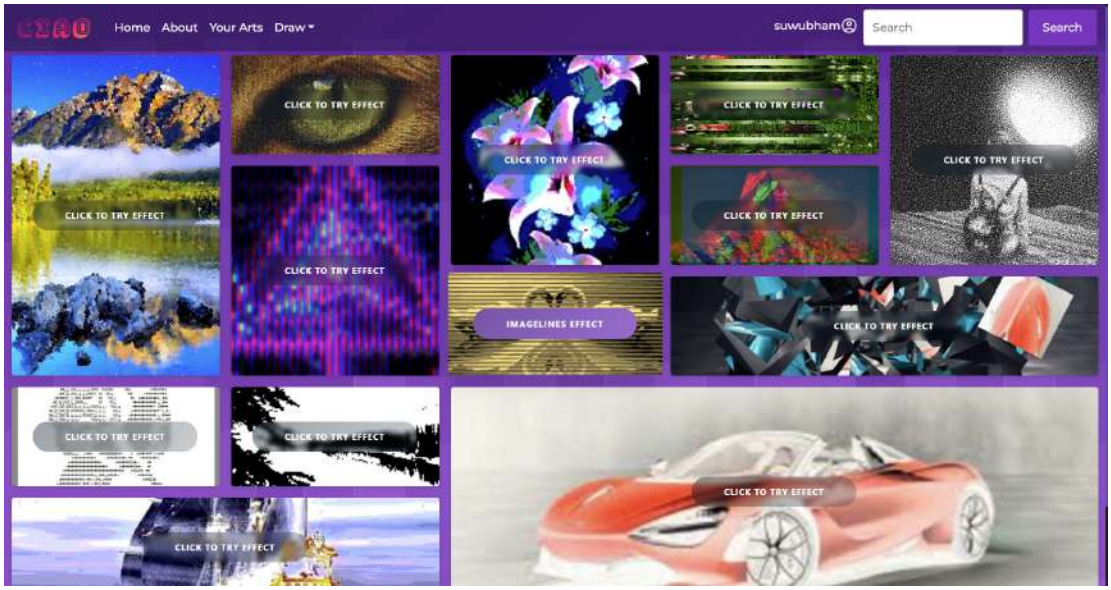

Fig 4.1.3.1: List of the effects for art from image

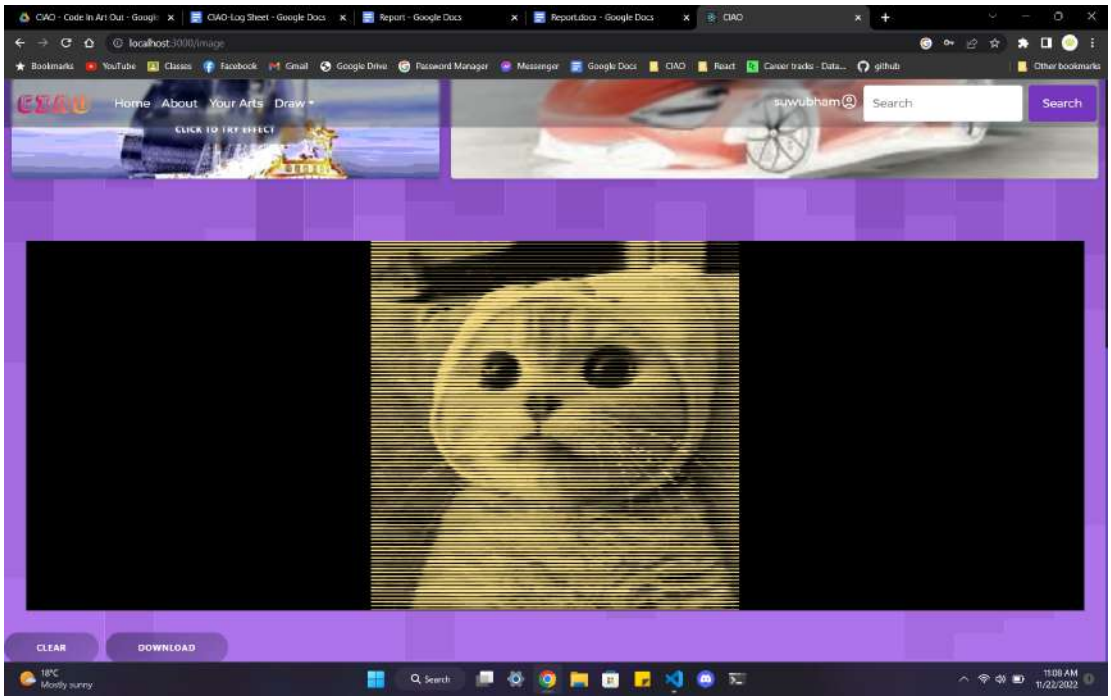

Fig 4.1.3.2: Adding image lines effect on the picture of a cat

4.1.4 Download, Share and save generated art

The user can download generated art from images and templates. They can rename it as per their preference. Furthermore, the art that they generate can also be saved in the cloud to view later. They can also share the url of the art that they wish they share.

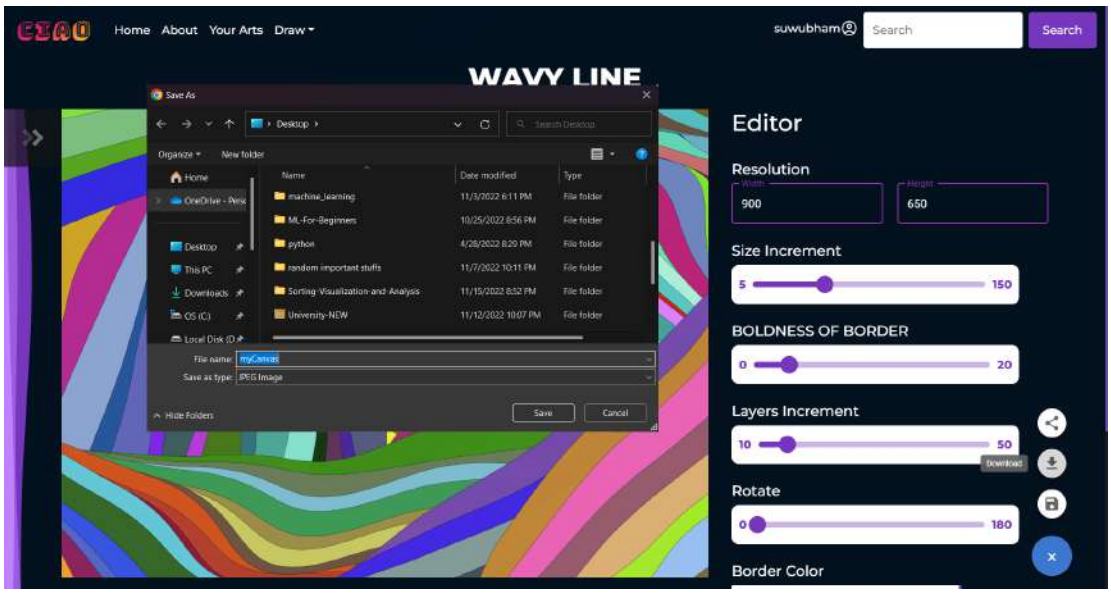

Fig 4.1.4.1: Download Generated art

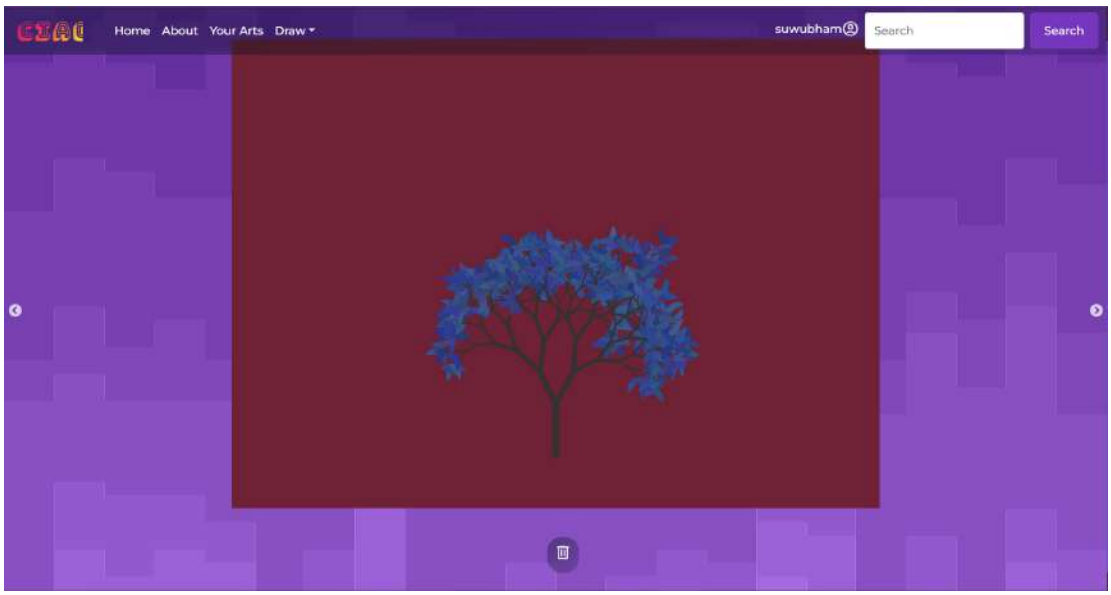

Fig: 4.1.4.2: Save Generated art

4.1.5 Add arts to favorites

Users can add arts to their favorites by tapping on the little heart icon below each art. The state of this page is saved in the database. They also view only their favorite arts only by selecting it in the filter dropdown.

Fig 4.1.5.1: No arts added to favorites

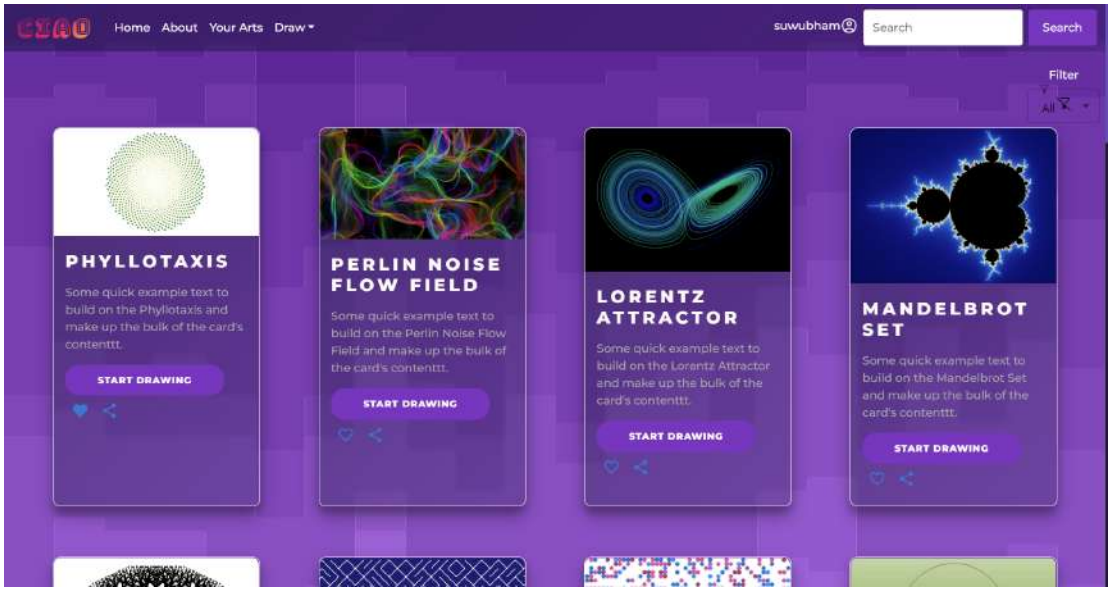

Fig 4.1.5.2: Adding arts to favorites

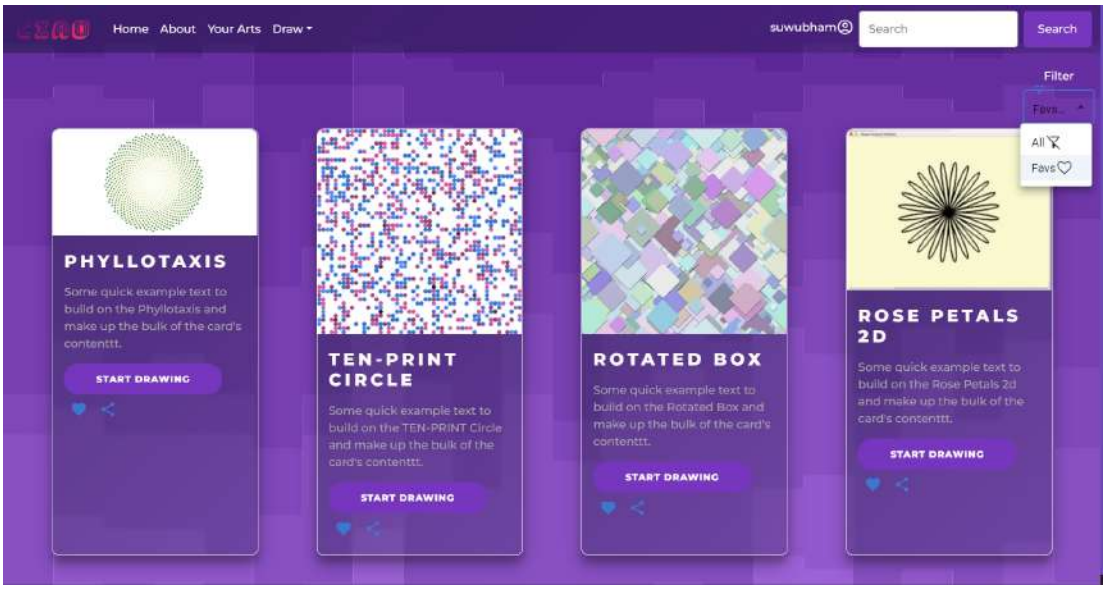

Fig 4.1.5.3: Viewing favorite arts only

4.2 Challenges

- Understanding MERN Stack for web development.
- Using p5.js to create generative art.
- Learning Git and Github for version control and group collaboration.
- Designing a responsive, interactive user interface using a color palette, template, editor tools, background, and layout.
- Backend and database integration using Node.js, Express.js, and mongoDB.

4.3 Achievements

- A beautiful, user-friendly interactive UI.
- Collection of 19 art from templates and 13 art from images for the user to choose from.
- Teamwork and project management skills.
- Created an open-source, free for all platform that is readily available on the internet.

Chapter 5: Conclusion and Recommendation

CIAO is an elegant web app packed with tons of features. Users can create art by choosing from 19 different templates that are fully customizable or from 13 different effects that can be applied to images. Creation of generative art promotes creative coding which can be a very fun way to learn about different programming concepts.

5.1 Limitations

In spite of the fact that the app is functional, some limitations remain. The major limitation in this app is not related to the functionality of the app but rather its UI. Some of the pages are not fully responsive so mobile users might find it hard to navigate through those pages. Furthermore, there is no forget password option, so users cannot recover their accounts if they forget their password.

5.2 Future enhancements

As for future enhancements, we are looking to utilize the search bar so that the user can easily search for the art instead of navigating through the entire grid of available arts. Also as discussed in limitations, the option to recover forgotten password while the user is trying to log in when they forget their password needs to be added.

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[1] Galantar, P. (2003). What is Generative Art? Complexity Theory as a Context for Art Theory.

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[8] Silk. (2022). [Web App]. http://weavesilk.com/

Appendix

The figure below is the Gantt chart which shows the time allocated for each part of the project. A minimum of 10 weeks to complete the project has been allocated.

Week Task	1	2	3	4	5	6	7	8	9	10
Research and Study										
Work division										
Frontend										
Backend										
Testing and Debugging										
Documentation										

Table: A.1: Gantt Chart